Examiner only

Option C – **The Physics of Sports**

9. (a) (i) Liam wishes to buy a baseball bat with a large moment of inertia and he chooses **bat B** from the following list in a catalogue. Evaluate whether he has chosen the correct bat. [2]

Bat	Mass / kg	Length / cm
A	1.00	85
B	1.10	82
C	1.05	85

....

....

....

....

(ii) Calculate the angular momentum of a baseball bat if it has a moment of inertia of 0.35 kg m^2 and is rotating at 63 rad s^{-1}. [2]

....

....

....

(b) (i) Calculate the torque that the pitcher must exert on a 145 g baseball with radius 3.70 cm to increase the baseball's spin from zero to 2400 revolutions per minute in a time of 0.20 s.
The moment of inertia of a baseball is given by the equation $I = \frac{2}{5}mr^2$. [4]

....

....

....

....

....

....

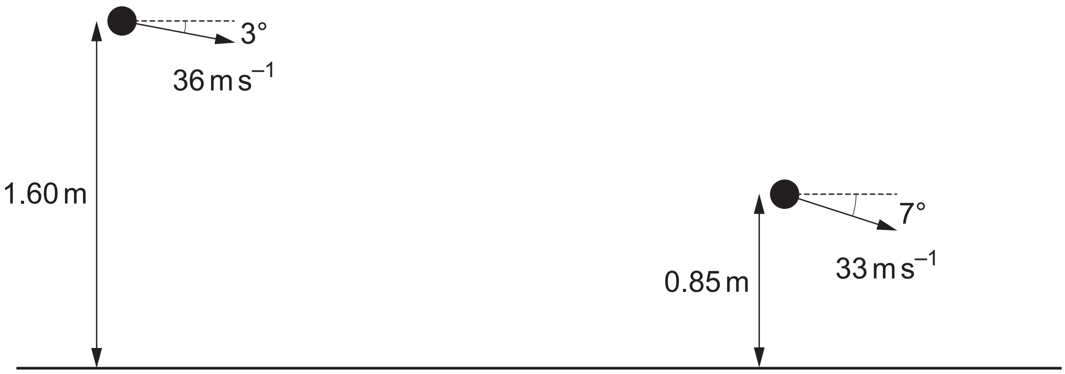
....

Examiner only

(ii) Calculate the rotational kinetic energy gained by the ball. [2]

(iii) By considering both the linear and rotational motion, explain why baseball players wear a glove on the hand that they use to catch the ball. [3]

(c) A fielder throws the ball towards a teammate. The ball is initially thrown from 1.60 m above the ground with a speed of $36\,\mathrm{m\,s^{-1}}$ at an angle of 3° below the horizontal. When the ball is caught it is 0.85 m above the ground and has a speed of $33\,\mathrm{m\,s^{-1}}$ at an angle of 7° below the horizontal.



(i) Explain how the information in the diagram shows that air resistance must be acting on the ball. [2]

Examiner only

(ii) Evaluate whether the vertical acceleration of the ball is equal to the acceleration due to gravity. [3]

....

....

....

....

....

....

(iii) Explain why a spinning ball may travel further in air than a non-spinning ball. [2]

....

....

....

20